Vidya Sagar Reddy Venna

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Signal Processing | Biosignals | Sensor Machine Learning

EDUCATION

University of Southern California Aug 2024 - May 2026
M.S., Electrical & Computer Engineering (Machine Learning & Data Science)

Indian Institute of Information Technology, Sri City Aug 2020 - May 2024
B.Tech (Honours), Electronics & Communication Engineering

Selected coursework: DSP, Statistical Signal Processing, Pattern Recognition, Embedded Systems

SKILLS

Tools: Python, C++, MATLAB, SciPy, PyTorch, scikit-learn, Docker
Modeling: Kalman filtering, PSID, CNNs, BiLSTMs, SVMs, mixed-effects models, cross-validation
Signal processing: sEMG, digital filtering, TKEO onset detection, MVIC normalization, MFCCs, log-mel spectrograms

EXPERIENCE

USC Information Sciences Institute Aug 2025 - Sep 2026
Graduate Student Researcher | Marina del Rey, CA

- **Knowledge extraction:** Built a pipeline converting scholarly PDFs into structured facts with groundedness checks.
- **Evaluation:** Retained 88.3% of full-text answer accuracy in a controlled LLM evaluation.

USC COOR Lab Jul 2025 - Present
Research Assistant, Clinical Biomechanics | Los Angeles, CA

- **sEMG processing:** Built a SciPy pipeline for a 54-patient rehabilitation study, with filtering, TKEO onset detection and MVIC normalization.
- **Statistical modeling:** Analyzed neuromuscular recovery using linear mixed-effects models and leave-one-out cross-validation; study submitted to APTA CSM 2027.

BrainSightAI Jan 2024 - Jun 2024
Machine Learning Intern | Bangalore, India

- **Diffusion MRI:** Validated a 17-step Docker pipeline with DIPY, MRtrix3, ANTs and FSL using hospital and public MRI data.
- **Denoising:** Compared Patch2Self and MPPCA; spatial-artifact review informed the production choice of MPPCA.

Human Movement Analysis, IIIT Sri City May 2022 - May 2024
Honours Student Researcher

- **Data collection:** Led collection and public release of sEMG datasets from 70 participants using Noraxon sensors, varying posture, arm position, pace and load.
- **Activity classification:** Trained CNN-BiLSTM models on 250 ms windows using filtering, rectification, normalization and feature extraction.

PROJECTS

Deep Audio Classification GitHub

- **Audio classification:** Collaboratively benchmarked MFCC/SVM, CNN, VGG-style and ResNet-18 models on 8,732 sound excerpts; log-mel with Mixup performed best.

State-Space Neural Decoding (coursework) GitHub

- **Neural decoding:** Compared Kalman filtering and PSID to decode movement offline from primate neural recordings.

Vibration Quadruped (EE579) GitHub

- **Embedded control:** Wrote C++ firmware for a 4-servo team prototype; timed vibration gave no traversal gain on the tested granular terrain and increased total energy use.

CuPy Open-Source Contributions Merged Aug 2026

- **Digital filters:** Fixed bilinear leading-zero handling (CuPy #10139); documented sliding_window_view (#10138).

PUBLICATIONS

S. Sreenivas, A. Turlapaty, S. Naidu, V. Sagar. Impact of Activity Pace and Arm Position on Classification of ADLs. IEEE EMBC 2024.

S. Naidu, A. Turlapaty, V. Sagar. Classification of Fine-ADL Using sEMG Signals Under Different Measurement Conditions. BIOSIGNALS 2024.

V. Sagar, A. C. Turlapaty, S. Naidu. Impact of Measurement Conditions on Classification of ADL using Surface EMG Signals. IEEE ISPA 2023 (first author).